



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI PERUMALAPPAN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

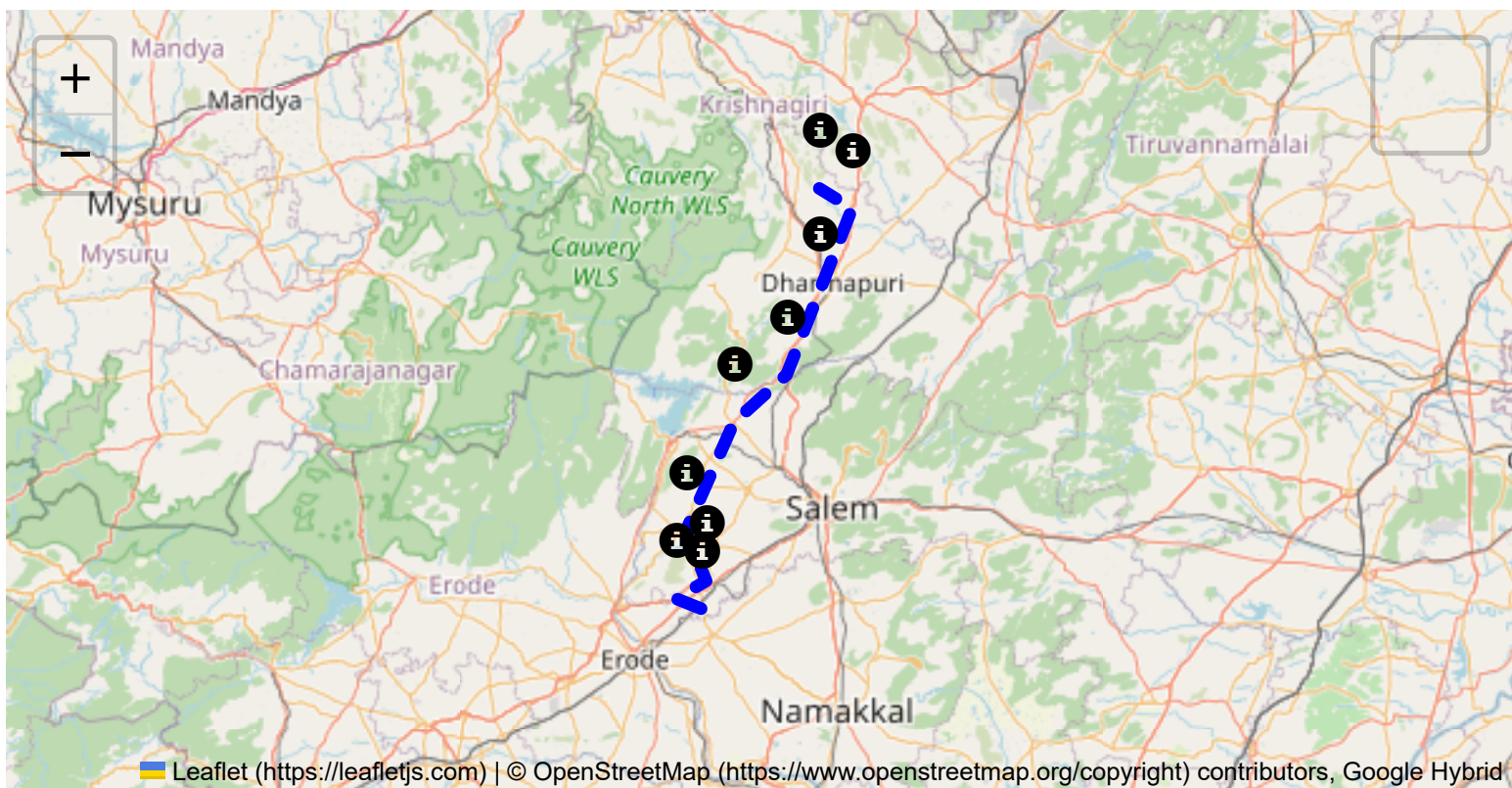
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

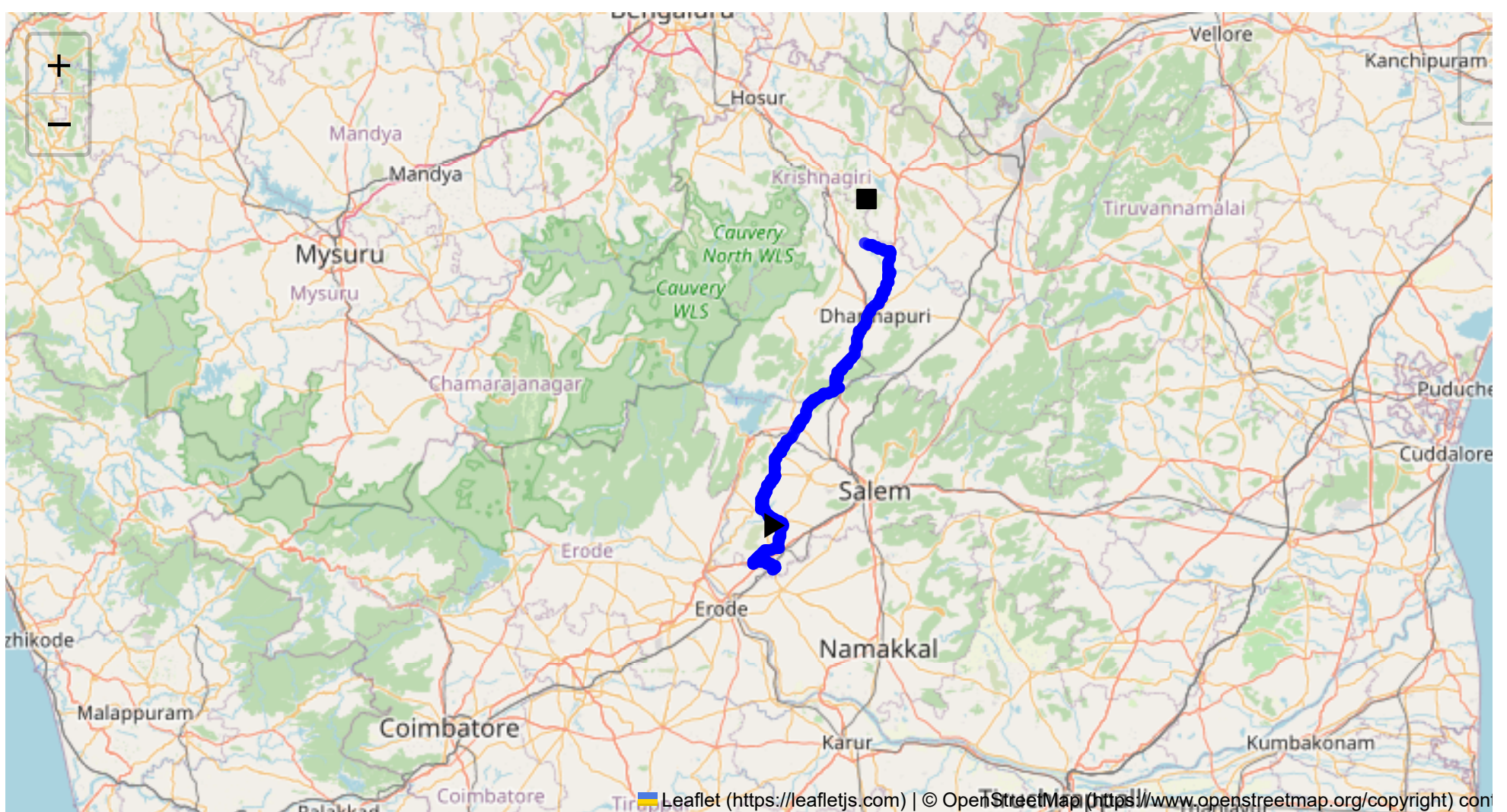
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 138.63 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.7 hours
Start: (11.4381, 77.8734)
End: (12.332181, 78.13153)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 138.63 kilometers within Tamil Nadu, India, and is usually completed in about 3 hours under normal conditions for heavy vehicles carrying hazardous materials. It connects various small towns and highways, including stretches on the Salem - Kochi Highway and other regional roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu often experiences tropical weather conditions. From June to September, the southwest monsoon can lead to heavy rains and potential flooding. During these months, roads might be slippery, increasing the risk of accidents, especially for heavy vehicles. December to March brings milder weather, whereas April to May can be very hot, causing potential overheating of engines.

3. Analysis of Traffic Patterns

- **Peak Hours:** Typically, mornings (8-10 AM) and evenings (5-8 PM) see higher traffic.
- **Congestion-Prone Areas:**
 - Salem - Kochi Highway can be congested, especially near urban areas like Salem.
 - The road segments around Thoppur are often busy due to nearby industrial activities.

4. Assessment of Road Quality and Infrastructure

The route includes major highways with generally good road quality. However, some regional roads might have uneven surfaces or potholes. During monsoon, these roads can deteriorate further.

5. Suggestions for Alternative Routes

In case of emergencies, potential alternate routes include:

- For bypassing a roadblock on the Salem - Kochi Highway, consider using state highways that run parallel to this section, although they might be longer and slower.
- Local roads connecting to national highways can serve as bypass paths around urban congestion.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu enforces strict guidelines for hazardous material transport, including permits and specific transit routes. Heavy vehicles are restricted from certain city areas during peak hours. There are also environmental regulations concerning emissions and noise.

7. Overview of Historical Incidents

Previous incidents involving heavy vehicles have primarily occurred due to brake failures on hilly terrains and speed-related accidents on highways. Not much data on hazardous material spills, but risks remain due to mixed traffic conditions.

8. Environmental Considerations and Sensitive Areas

The route passes near agricultural areas, particularly around Thoppur and Adilam, which could be sensitive to chemical spills. Forested areas near Thoppur also require caution to avoid ecological damage.

9. Analysis of Communication Coverage

Major highway segments have good communication coverage. However, certain remote portions between Thoppur and Thadangam might experience connectivity issues due to terrain and limited infrastructure.

10. Estimated Emergency Response Times

Emergency response times vary:

- Urban areas like Salem: Approximately 15-20 minutes.
- Remote areas, particularly the sections between Thoppur and Thadangam: Can exceed 45 minutes due to distance and accessibility.

12. Overall Summary of Risk Assessment

The route presents moderate risk levels for transporting hazardous materials, primarily concerning weather and traffic. Precautions are advised during the monsoon season and peak traffic times. While infrastructure is largely adequate, alternate route knowledge and emergency contacts are crucial for handling contingencies effectively. Enhanced training for drivers on local regulations and environmental awareness can further mitigate risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58303, 77.85827	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	U-Turn	High	11.9367046, 78.0560593	10 KM/Hr
3	Turn	High	11.43961, 77.87341	15 KM/Hr
4	Turn	High	11.44024, 77.87528	15 KM/Hr
5	Turn	High	11.44898, 77.87410	15 KM/Hr
6	Turn	High	11.45352, 77.85698	15 KM/Hr
7	Turn	Medium	11.45764, 77.85219	30 KM/Hr
8	Turn	High	11.46303, 77.84945	15 KM/Hr
9	Turn	Medium	11.45543, 77.81740	30 KM/Hr
10	Turn	High	11.45584, 77.81695	15 KM/Hr
11	Turn	Medium	11.45656, 77.81709	30 KM/Hr
12	Turn	High	11.49383, 77.88442	15 KM/Hr
13	Turn	High	11.56083, 77.89921	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	Medium	11.56179, 77.89726	30 KM/Hr
15	Turn	High	11.56459, 77.89593	15 KM/Hr
16	Turn	Medium	11.56584, 77.89147	30 KM/Hr
17	Turn	High	11.56641, 77.89130	15 KM/Hr
18	Turn	Medium	11.56764, 77.88847	30 KM/Hr
19	Turn	High	11.56925, 77.88785	15 KM/Hr
20	Turn	Medium	11.58709, 77.85824	30 KM/Hr
21	Turn	Medium	11.60960, 77.84159	30 KM/Hr
22	Turn	Medium	11.64055, 77.84600	30 KM/Hr
23	Turn	Medium	11.64173, 77.84878	30 KM/Hr
24	Turn	Medium	11.64600, 77.85144	30 KM/Hr
25	Turn	Medium	11.65232, 77.85184	30 KM/Hr
26	Turn	Medium	11.66451, 77.85595	30 KM/Hr
27	Turn	Medium	11.66583, 77.85601	30 KM/Hr
28	Turn	Medium	11.67289, 77.86017	30 KM/Hr
29	Turn	High	11.68488, 77.86889	15 KM/Hr
30	Turn	Medium	11.68458, 77.87219	30 KM/Hr
31	Turn	High	11.68486, 77.87288	15 KM/Hr
32	Turn	High	11.69437, 77.87569	15 KM/Hr
33	Turn	Medium	11.69491, 77.87546	30 KM/Hr
34	Turn	High	11.69551, 77.87442	15 KM/Hr
35	Turn	Medium	11.75989, 77.89002	30 KM/Hr
36	Turn	Medium	11.76018, 77.89078	30 KM/Hr
37	Turn	High	11.75964, 77.89200	15 KM/Hr
38	Turn	Medium	11.83316, 77.94421	30 KM/Hr
39	Turn	Medium	11.83368, 77.94485	30 KM/Hr
40	Turn	Medium	11.84163, 77.94718	30 KM/Hr
41	Turn	Medium	11.89734, 77.98068	30 KM/Hr
42	Turn	Medium	11.89985, 77.98191	30 KM/Hr
43	Turn	Medium	11.91892, 78.02607	30 KM/Hr
44	Turn	Medium	11.93612, 78.05384	30 KM/Hr
45	Turn	Medium	11.93673, 78.05414	30 KM/Hr
46	Turn	Medium	11.93695, 78.05464	30 KM/Hr
47	Blind Spot	Blind Spot	11.93670, 78.05606	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
48	Turn	Medium	11.93694, 78.05425	30 KM/Hr
49	Turn	Medium	11.95531, 78.04698	30 KM/Hr
50	Turn	High	11.95803, 78.04964	15 KM/Hr
51	Turn	Medium	11.96068, 78.04960	30 KM/Hr
52	Turn	High	11.96370, 78.05142	15 KM/Hr
53	Turn	High	11.96542, 78.04938	15 KM/Hr
54	Turn	High	11.96650, 78.04927	15 KM/Hr
55	Turn	Medium	12.22410, 78.19125	30 KM/Hr
56	Turn	Medium	12.24670, 78.19551	30 KM/Hr
57	Turn	Medium	12.24996, 78.19933	30 KM/Hr
58	Turn	Medium	12.25264, 78.20041	30 KM/Hr
59	Turn	High	12.30921, 78.20317	15 KM/Hr
60	Turn	Medium	12.30953, 78.20254	30 KM/Hr
61	Turn	Medium	12.31038, 78.20211	30 KM/Hr
62	Turn	Medium	12.31086, 78.20127	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
2	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
3	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
4	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
5	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
6	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium
8	hospital	Vasu Speciality Hospital	11.8331097, 77.943169	30 km/h	Medium
9	hospital	Rajamanickam Hospital	11.8332408, 77.9431081	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	hospital	Child Hospital	11.833001, 77.943415	30 km/h	Medium
11	hospital	Naveen Nursing Home	11.8335355, 77.9437441	30 km/h	Medium
12	hospital	Sri Sakthi Hospital	11.835714, 77.9451014	30 km/h	Medium
13	clinic	Sri Krishna Clinic	11.835443, 77.9450587	30 km/h	Medium
14	hospital	Selvakumar Hospital	11.836215, 77.945842	30 km/h	Medium
15	clinic	SK Clinic	11.83619, 77.9457752	30 km/h	Medium
16	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
18	pharmacy	Appolo	12.3088621, 78.2039069	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	college	Kavery Educational Institutions	11.8255319, 77.9342256	30 km/h	Medium
17	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58303, 77.85827



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60348, 77.83967



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.9367046, 78.0560593



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45584, 77.81695



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45656, 77.81709



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49383, 77.88442



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.56083, 77.89921



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56179, 77.89726



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.56459, 77.89593



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56584, 77.89147



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.56641, 77.89130



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56764, 77.88847



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.56925, 77.88785



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58709, 77.85824



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60960, 77.84159



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64055, 77.84600



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64173, 77.84878



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64600, 77.85144



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65232, 77.85184



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66451, 77.85595



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66583, 77.85601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67289, 77.86017



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68488, 77.86889



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68458, 77.87219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68486, 77.87288



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69437, 77.87569



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69491, 77.87546



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69551, 77.87442



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.75989, 77.89002



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76018, 77.89078



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.75964, 77.89200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83316, 77.94421



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83368, 77.94485



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84163, 77.94718



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.89734, 77.98068



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.89985, 77.98191



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91892, 78.02607



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93612, 78.05384



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93673, 78.05414



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93695, 78.05464



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.93670, 78.05606



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93694, 78.05425



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95803, 78.04964



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96068, 78.04960



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22410, 78.19125



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.24670, 78.19551



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.24996, 78.19933



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25264, 78.20041



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.30921, 78.20317



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.30953, 78.20254



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.31038, 78.20211



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.31086, 78.20127
